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Assignment 2: Appliance Energy Forecasting Report

Introduction

Appliance-level load is harder to forecast than whole-building demand. Aggregate demand is smoothed by averaging over many devices, whereas a single household's appliance use is dominated by discrete switching events, a kettle or washing machine turning on produces a step change no smooth model reproduces. Candanedo et al. [1], who released the dataset used here, reported that even their best regression models left much of the variance unexplained.

We forecast appliance use 24 hours ahead and ask a deliberately sceptical question: how much accuracy is available from very simple methods, and what does each layer of complexity actually buy? Five benchmarks, a SARIMAX model, a feature-based regressor and a pretrained foundation model are compared on the same test period and horizon. The framing follows Hyndman and Athanasopoulos [2] in anchoring everything to benchmarks, and echoes the M4 competition finding that simple methods stay competitive more often than expected [3]. Two results really shaped the analysis, firstly, the advanced models beat the best benchmark by less than expected, and secondly, the sensor and weather covariates the brief asks for made the feature model **worse**.

Data and preprocessing

The dataset [1] holds 19,735 observations at 10-minute resolution from one low-energy Belgian house between 11 January and 27 May 2016, with a lighting meter, nine paired indoor temperature, humidity sensors, and airport weather data alongside the target.

We aggregated to hourly means giving 3,290 observations. Hourly data makes seasonal ARIMA tractable, a seasonal period of 144 at 10-minute resolution produces an unwieldy state space, whereas 24 is routine and smooths some switching noise, at the cost of averaging away genuinely sub-hourly behaviour. Gaps from resampling were time-interpolated with no target values were missing afterwards.

The target is strongly right-skewed (mean 97.8 Wh, median 63.3 Wh, skewness 2.39, excess kurtosis 6.33). 70% of hours fall below 100 Wh while the maximum reaches 608 Wh. That shape drives much of what follows since a model can fit the quiet majority well and still be badly wrong about the few hours that dominate squared error. We did not log-transform, so MAE and RMSE stay interpretable in watt-hours and their disagreement stays visible.

Exploratory analysis and stationarity

Demand is lowest overnight, climbs through the morning and peaks in the early evening, with a weaker weekly component lifting weekends. The ACF decays slowly with clear spikes at lags 24, 48 and 72 and a smaller bump near 168.

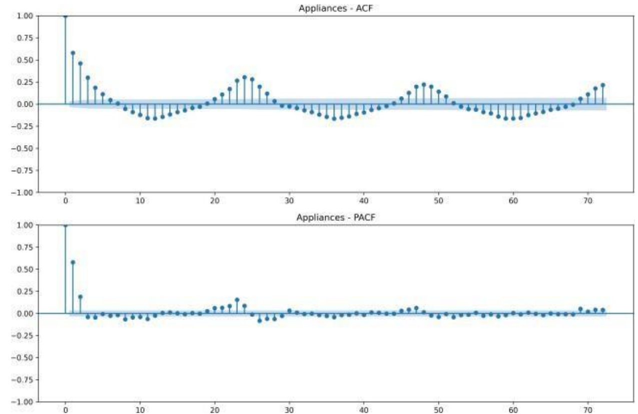

Figure 1. ACF and PACF of hourly appliance use

The Augmented Dickey–Fuller test [4] rejects a unit root on the raw series (−8.95, $p < 0.001$), which is initially counter-intuitive given the visible cycles. Appliance load is physically bounded and mean-reverting, so it cannot wander like a random walk and is stationary in the ADF sense, that says nothing about periodic structure. Differencing the level would have risked over-differencing and inflated the MA order, while the daily cycle still needs modelling explicitly. Hence $d = 0$ with a seasonal difference $D = 1$, not the reverse.

Forecasting design

The horizon is 24 hours, the test period the final 14 days (336 hours), and training uses the preceding 2,954 hours. Models are scored with MAE, RMSE, MASE and mean bias, MASE is scaled by the in-sample daily seasonal-naive error, so below one beats that reference [5]. One decision does most of the work. Forecasting all 336 test steps in one pass would measure 336-step accuracy, a different and much harder problem than the one posed. We use rolling-origin evaluation [6]: forecast 24 hours, reveal the realised values, advance the origin, repeat for 14 blocks. Every prediction is then at most 24 steps ahead and no model sees the block it predicts. This is not cosmetic evaluated over 336 steps the naive method scores MASE 4.69, but at the stated horizon it scores 1.60. The first figure penalizes it for a task nobody set.

Three safeguards against leakage were applied, lag and rolling features use only target values from at least 24 hours earlier (Section 7), model and feature selection used a validation block preceding the test period, never the test set and no scaler was fitted across the train/test boundary, which tree models make unnecessary anyway.

Benchmark models

Five benchmarks were fitted. The weekly seasonal naive is strongest (MASE 0.813), the daily version second (0.904); both beat the mean (0.941) and are roughly twice as accurate as naive and drift (1.601, 1.606). That repeating last week beats repeating yesterday says the household's routine is organised around the week rather than the day, the same hour on the same weekday is a better guide, presumably because weekday and weekend behaviour differ. That both seasonal methods crush naive and drift says the series carries almost no exploitable trend or persistence beyond the cycle. Everything else must beat MASE 0.813 to justify itself.

SARIMAX

Order selection followed AIC [8] over the required grid $p \in [0,6]$, $d \in [0,2]$, $q \in [0,6]$, 147 combinations. Fitting all 147 with a period-24 seasonal component was not feasible since one such fit takes minutes once the seasonal state space gains ~24 dimensions. We screened all 147 non-seasonal orders then refined the leaders against a small seasonal grid and the full grid is searched, in two stages.

Two problems changed the outcome. Only 89 of 147 screening fits converged and none of the top ten did, one returning an AIC above 200,000, so the leaders were refitted at a higher iteration cap before any choice was made. Second, with no seasonal term available the screening stage compensated by choosing high AR and MA orders that imitate the daily cycle, so we also carried several deliberately parsimonious orders into the seasonal stage. This proved decisive: SARIMA(2,0,2)(0,1,1,24) beat SARIMA(6,0,3)(1,1,1,24) by 451 AIC points, and taking the screening ranking at face value would have selected an over-parameterised model that does not converge. The selected model scores AIC 32,198 against 32,967 for the best converged non-seasonal model on the same 2,954 observations, so the seasonal term is worth 769 points. The coefficients explain why the seasonal MA term is overwhelmingly significant (−0.95, $|z| \sim 170$) while ar.L2, ma.L1 and ma.L2 are not significant at 5%.

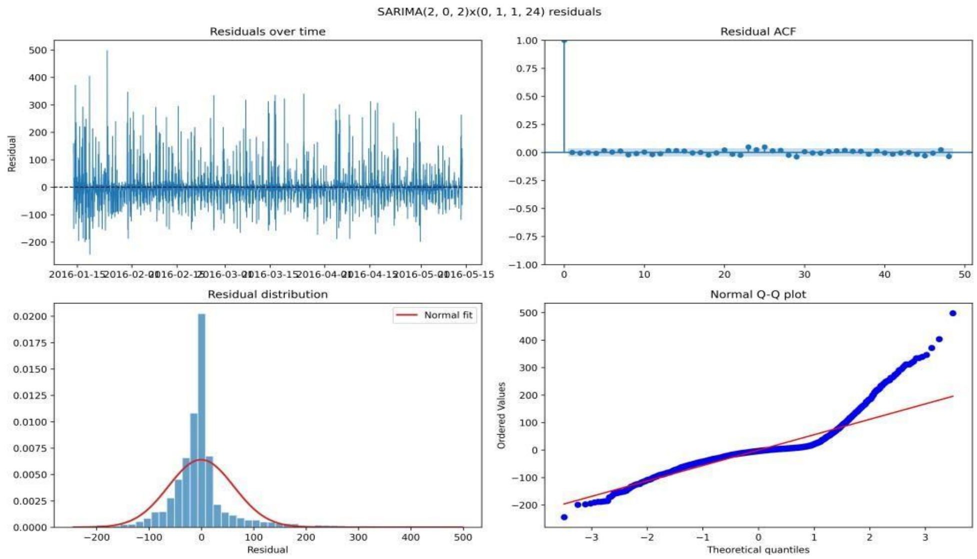

Figure 2. Residual diagnostics for SARIMA(2,0,2)(0,1,1,24). The ACF is essentially clean.

Diagnostics give two verdicts. Ljung–Box [9] returns $p = 0.41$, so we cannot reject uncorrelated residuals — the model absorbed the dependence it was built for. Normality is rejected decisively (Jarque–Bera $p < 0.001$; skewness 1.88, kurtosis 11.19), with a sharp right tail, exactly what one expects when demand sits low then spikes. This limits the intervals more than the point forecasts, since the bands are symmetric while the errors are not. Adding outdoor weather as exogenous regressors did not help: every coefficient was insignificant (p between 0.18 and 0.86), AIC rose to 32,206, and rolling accuracy was marginally worse. Once the daily cycle is modelled, weather adds almost nothing at this horizon. The cycle already encodes the occupancy routine driving demand.

Feature-based model

We built a 50-column table in six groups. calendar features from the timestamp, lagged target values, rolling summaries, nine indoor sensors plus derived house-level aggregates, outdoor weather, and the lighting meter. Hour and day-of-week were sine/cosine encoded so the model treats them as cyclical.

The horizon constrains which lags are admissible, and this is easy to get wrong. At origin o we predict o+1 to o+24; a lag-1 feature on the row at o+24 would need the target at o+23, inside the block being predicted. Short lags are valid one step ahead but leak at 24, as do rolling windows shifted by a single step. We admit only lags ≥ 24 and shift rolling windows by the full horizon. The effect is large: lag-1 correlates with the target at 0.58 against 0.31 for lag-24, and a one-step-shifted rolling mean at 0.18 against 0.004 for the safe version. A model given the leaky features would have scored well and been impossible to deploy.

Feature groups were added cumulatively and scored on a validation block covering the 14 days before the test period, so no selection decision touched the test data.

Feature groups (cumulative)	**n**	**Val. MAE**	**Val. MASE**
calendar	7	39.59	0.741
lag	**13**	**35.50**	**0.664**
rolling	19	47.37	0.887
house (lights)	20	45.40	0.850
indoor sensors	44	78.02	1.460
outdoor weather	50	78.14	1.463

Table 1. Cumulative feature-group ablation on the validation block.

Calendar and lag features together give the best score; everything added afterwards makes the model worse, and the indoor sensors roughly double the error. Two mechanisms are at work. The sensor and weather readings are weakly related to appliance use once the cycle is captured, the strongest sensor correlation is 0.17, so 31 extra columns mostly supply noise. More importantly, indoor temperature and humidity drift across the season and both validation and test periods sit at the end of a warming spring, so the model learns a relationship between absolute temperature and demand that no longer holds a fortnight later. The nine sensors are also highly collinear (mean pairwise 0.79, maximum 0.95), adding far less independent information than their count suggests. On the selected 13 features a random forest [10] beat XGBoost [11] and histogram gradient boosting on validation MASE (0.628, 0.664, 0.687). We put this down to sample size: with 2,427 rows and 13 features, the variance reduction of bagging suits the problem better than sequential boosting, which had more capacity to overfit than was needed. The final model was refitted on training plus validation, then scored once on the test set.

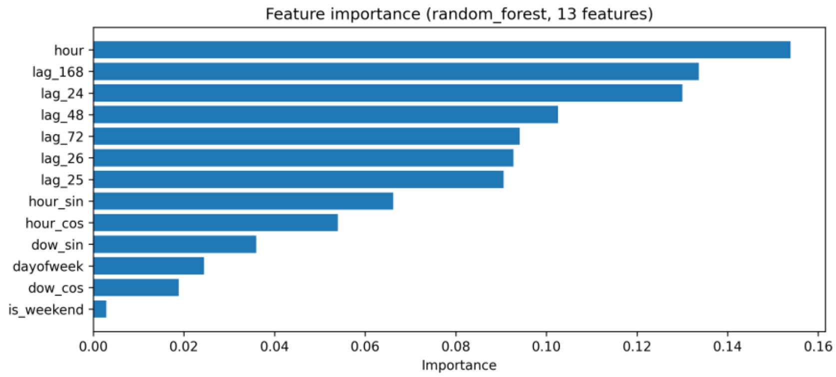

Figure 3. Random-forest feature importances. Hour of day and the daily and weekly lags dominate.

Hour of day leads, followed by lag-168 and lag-24, the same weekly-then-daily ordering the benchmarks showed. On the full 50-feature model the indoor sensors attract the largest share of importance despite the ablation showing they damage accuracy. Split-based importance measures how often a feature was used, not whether it helped, and is biased towards correlated predictors [12]. Where importance and ablation disagree, the ablation is better evidence.

Foundation model

We used Chronos-Bolt-small, a 47.7M-parameter model pretrained on a large corpus of time series. TimeGPT [13] was rejected because it needs an API key and a network call per forecast, preventing the pipeline running from a fresh clone; TimesFM is a comparable open alternative. Chronos was applied strictly **zero-shot**: published weights unchanged, no fitting or fine-tuning, its only exposure to the dataset being the context window at prediction time. It is also **univariate**, so it received none of the covariates from Section 7 and works from strictly less information than the other two advanced models.

It produced the best MAE and MASE of any model (34.27, 0.641) in under six seconds for all fourteen blocks, with 80% intervals covering 83.6% of realised values. We report an 80% rather than 95% interval deliberately: Chronos-Bolt is trained only on quantiles 0.1–0.9, and asking for 0.05/0.95 silently returns the 0.1/0.9 band, which would be an 80% interval mislabelled as 95%. SARIMAX has no such restriction. This is a real limitation of the foundation model. Scaling helps weakly, tiny (8.7M) scores MASE 0.651, small (47.7M) 0.641, base (205.3M) 0.623 — about 4% for a 24-fold parameter increase. We kept small as default because base is roughly a 1 GB download for 0.02 MASE, a deployment judgement rather than a statistical one.

Results and error analysis

Model	MAE	RMSE	MASE	Bias	vs benchmark
Foundation (Chronos-Bolt-small)	**34.27**	**68.09**	**0.641**	**−16.57**	**+21.2%**

Feature-based (random forest)	35.38	62.03	0.662	−2.37	+18.6%
SARIMAX(2,0,2)(0,1,1,24)	36.26	64.74	0.679	−6.33	+16.6%
Weekly seasonal naive	**43.46**	**81.41**	**0.813**	**−13.16**	**—**
Daily seasonal naive	48.31	85.57	0.904	+1.75	−11.2%
Mean	50.26	74.94	0.941	−3.29	−15.7%
Naive	85.55	110.39	1.601	+50.98	−96.9%
Drift	85.80	110.68	1.606	+51.37	−97.4%

Table 2. Accuracy over the 14-day test period.

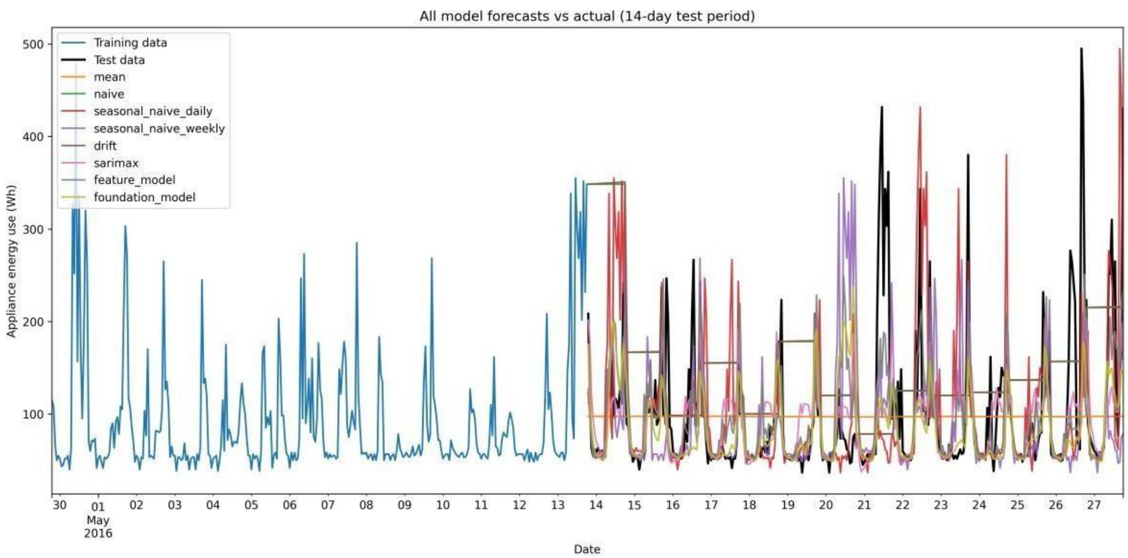

Figure 4. All model forecasts against realised demand over the test period.

All three advanced models clear the strongest benchmark by 17–21% on MASE, so the complexity is justified on accuracy. But they sit within three percentage points of each other and the ranking is inconsistent across metrics: Chronos wins on MAE and MASE, the random forest on RMSE. That disagreement is the most informative result in the table.

The error distribution explains it. Chronos has by far the lowest median absolute error (9.8 Wh against 14.7 and 15.9) but the worst 90th percentile of the three (108.5 against 91.9 and 80.0). It models ordinary hours better and misses evening spikes by more. RMSE squares errors and so rewards avoiding large misses; MAE rewards being usually close. Neither is wrong, and which should govern depends on the application: daily energy totals favour the low median, sizing a supply to cover peaks favours the controlled tail.

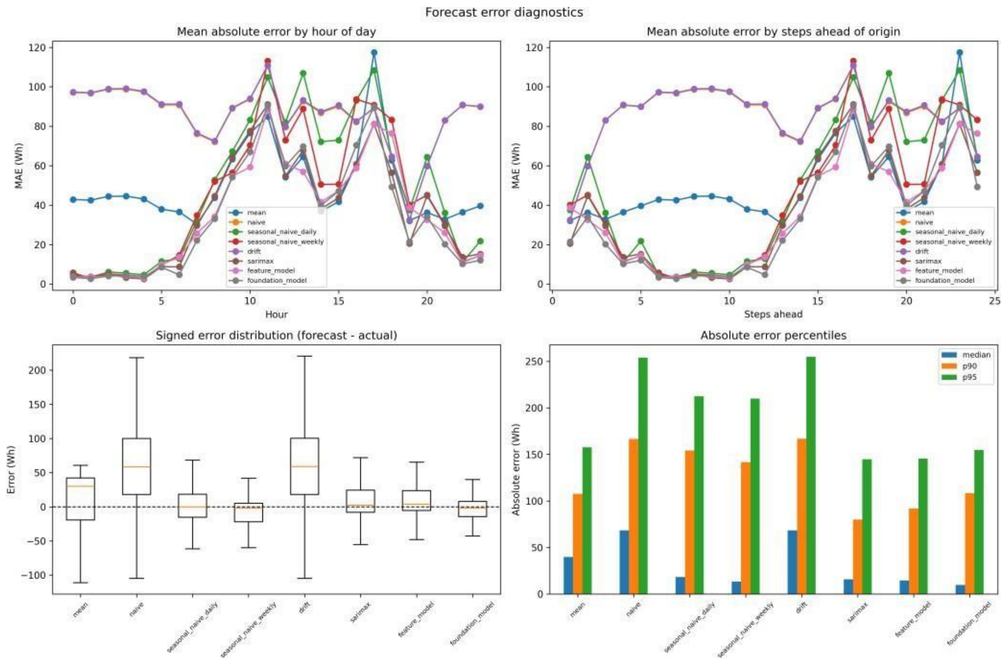

Figure 5. Error diagnostics: absolute error by hour of day and by position in the forecast block (top), signed error distributions and error percentiles (bottom).

One caveat on the step-position panel, which we only noticed after building it. The test period is fourteen aligned 24-hour blocks, so step position and hour of day are the same variable: step 1 is always 19:00, step 24 always 18:00. It cannot separate forecast distance from time-of-day difficulty, and a rising profile must not be read as the former. Comparing models at a fixed step is still valid, since all predict the same hour, and step 24 is cleanest because every model is then a full day ahead. There the order is Chronos 49.2, SARIMAX 56.4, random forest 76.3, weekly seasonal naive 83.2 — SARIMAX and the forest have swapped relative to Table 2. The rolling scheme hands SARIMAX and Chronos short-horizon steps at 23 of 24 positions, while the forest predicts every row a day ahead. So the aggregate is mildly confounded, and the step-24 figures rest on 14 observations each: SARIMAX and the feature model are hard to separate, and only Chronos's lead survives both views.

Bias and residual structure complete the picture. Chronos has the largest negative bias (−16.6 Wh), under-forecasting as a model hedging toward the typical level and clipping peaks would; the forest is nearly unbiased (−2.4 Wh). Error autocorrelation at lag 24 is essentially zero for the forest (−0.03) and small for SARIMAX (0.14), which leaves nothing at lag 168 either — both absorbed the cycle they were built for. The mean forecast retains 0.38 and 0.35 at those lags, unexploited structure the others are using. The dominant failure is shared: every model reproduces the daily shape but under-predicts evening peaks, and the hardest hours are the same throughout. That is the switching behaviour described in [1], which no smooth conditional-mean model captures.

Assignment Part 9 Answers

9.1 Which benchmark is strongest, and what does it tell us?

The weekly seasonal naive (MASE 0.813), ahead of the daily version (0.904), with naive and drift far behind (1.601, 1.606). Appliance use is governed by repeating human routine rather than level or momentum, and that routine is weekly, the same hour on the same weekday predicts better than the same hour yesterday, as expected if weekday and weekend behaviour differ. The failure of naive and drift shows little exploitable persistence or trend once the cycle is removed.

9.2 Does SARIMAX improve on the strongest seasonal benchmark?

Yes, by 16.6% on MASE. Daily seasonality and short-term autocorrelation are captured well. The seasonal MA term is significant, worth 769 AIC points, and Ljung–Box cannot reject uncorrelated residuals ($p = 0.41$). Weekly seasonality is handled less well, since we fitted only a period-24 component and the weekly benchmark's success suggests a second period was warranted. Exogenous variables are not adequately captured because there is little to capture, all five weather coefficients were insignificant and AIC worsened. The clearest inadequacy is distributional rather than dynamic, strongly non-normal residuals mean the symmetric intervals misrepresent the risk of an evening spike.

9.3 Does the feature-based model improve as features are added?

Only briefly, then it deteriorates sharply (Table 1). Calendar alone gives validation MASE 0.741, adding lags improves it to 0.664, and adding rolling windows, the lighting meter, indoor sensors and weather degrades it to 1.463. The useful groups are **time-of-day and lagged demand**. The harmful ones are sensor and weather. We attribute this to weak marginal signal, heavy collinearity among the nine indoor sensors, and seasonal drift in absolute temperature and humidity that does not generalise across a fortnight. Rolling features are weak because, once shifted a full 24 hours, a 24-hour average says little about a specific future hour of a spiky series.

9.4 Does the foundation model outperform the rest, and is it justified?

It gives the best MAE and MASE (34.27, 0.641), beating everything, and keeps its lead at the equal-horizon step-24 comparison. It does not win on RMSE, where it is weakest of the three, and it has the largest bias. Whether that justifies the complexity depends on how complexity is counted. Against SARIMAX's six coefficients, Chronos brings 47.7M parameters, a PyTorch dependency and a weights download for a 5.6% MASE gain and a 5% RMSE loss — a poor trade. Against the random forest it requires **no training, no feature engineering and no retraining schedule**, a real operational saving. Our view: the accuracy gain alone does not justify it, but accuracy combined with zero setup cost might, where no modelling expertise is available.

9.5 Which variables are known at the forecast origin?

Only 19 of 50 features (38%). Calendar features are known indefinitely ahead; lag and rolling features are known provided they respect the 24-hour rule. The other 31 — nine indoor temperatures, nine humidities, six derived aggregates, six weather variables and the lighting meter — are measurements whose future values do not exist when the forecast is made. A model fed their realised test values produces a **conditional forecast**, conditional on knowing future weather and sensor readings, not a true operational one. Our exogenous SARIMAX is exactly that and is reported as such. The recommended feature model avoids the issue: all 13 selected features are in the known-at-origin set, so the honest model and the accurate model coincide.

9.6 Which model would we recommend in practice?

The **feature-based random forest**, despite Chronos scoring marginally better on MAE. On accuracy the three are close, and the random forest has the best RMSE and smallest bias, which matter for sizing supply against peaks. On interpretability it is auditable — Figure 3 explains it in one plot — where Chronos is opaque. On uncertainty SARIMAX is strongest (analytic intervals at any level) and Chronos is capped at 80%. On computational cost the forest trains in seconds on 13 features with no GPU or large dependency. On deployment it uses only covariates genuinely known at the origin, making it a true forecast, and retrains cheaply as the household's routine drifts. If no modelling capability were available, zero-shot Chronos would be the pragmatic alternative; if calibrated uncertainty were the priority, SARIMAX would be.

Limitations and future work

The most useful extension would decouple horizon from time of day by advancing the origin hourly rather than daily. As built, our step-position diagnostic cannot separate the two; 336 overlapping origins would resolve it at modest cost.

- **Weekly seasonality in SARIMAX.** We fitted one seasonal period. Given the weekly benchmark's strength, a multiple-seasonality model such as TBATS, or Fourier terms at both periods, is a natural next step [2].

- **Modelling the spikes.** All models under-predict evening peaks. Treating demand as a baseline plus a switching process, or forecasting quantiles directly rather than the conditional mean, addresses the failure mode rather than working around it.
- **Single house, single season.** 137 days of one dwelling. Nothing establishes that the ranking transfers to other households or winter heating loads; the building-energy literature reports substantial heterogeneity.
- **Statistical significance.** Two to three MASE points over 336 hours may not exceed sampling noise. A Diebold–Mariano test or blocked resampling would establish whether the ranking is real.

Conclusion

Forecasting appliance load a day ahead is largely a matter of learning a household's weekly routine. A weekly seasonal naive reaches MASE 0.813 with no fitting at all, and the three advanced models improve on it by 17–21% real, but modest against the difference in complexity. The zero-shot foundation model gives the best typical accuracy, the random forest the best tail behaviour and smallest bias, SARIMAX the most trustworthy uncertainty. We recommend the random forest, which combines competitive accuracy with interpretability, negligible cost, and reliance only on covariates that would actually be known when the forecast is made.

The two most instructive results were negative. Adding the sensor and weather covariates the brief invites made the feature model substantially worse, and the exogenous SARIMAX failed to improve on the target-only version even when handed future weather it could not have known. Availability, collinearity and seasonal drift matter as much as raw correlation when choosing covariates, and here, the model that is honest about what it can know turned out to be the accurate one too.

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